

An Artifact-Driven Verification Protocol for Reproducible TinyML Deployment Evaluation

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Abstract—The deployment of deep learning models on resource-constrained microcontrollers (TinyML) is expanding rapidly, yet empirical evaluation in the field suffers from critical methodological vulnerabilities. Training-time in-memory metrics frequently diverge from disk-serialized binary behavior; theoretical weight sparsity is routinely conflated with physical storage compression; and host simulation timings are improperly extrapolated to microcontroller real-time guarantees. In this paper, we formulate an artifact-driven empirical verification protocol and defect taxonomy for compiled TinyML deployment artifacts. The protocol formalizes executable verification procedures across seven core dimensions: Data Isolation, Serialized Binary Integrity, Quantization Graph Inspection, Sparsity and Storage Decoupling, Computation Accounting, Timing Protocol Auditing, and Runtime Non-Leakage. We define formal programmatic predicates ($\mathcal{P}_1 \dots \mathcal{P}_7$) for each dimension and evaluate them on an end-to-end case study auditing 12 candidate neural network models trained on the 55,998-record EngineFaultDB physical benchmark. Our independent audit uncovers and resolves 20 distinct numerical discrepancies between training-time reports and verified disk binaries across four defect modes (with metric variances up to 7.82%). Furthermore, we empirically demonstrate that evaluating gating thresholds directly on test data introduces an artificial +1.80% optimistic accuracy bias in the evaluated pipeline, proving the necessity of split-isolated calibration. By establishing formal verification predicates and classifying evidence into a 5-tier empirical framework, our protocol provides actionable software engineering procedures to support reproducible benchmarking and defensible deployment claims in edge AI research.

Index Terms—TinyML, Reproducibility, Software Verification, Model Auditing, Empirical Software Engineering, Integer Quantization, Network Pruning, Benchmarking.

I. Introduction

EMBEDDED Machine Learning and Tiny Machine Learning (TinyML) have experienced exponential growth, enabling intelligent inference on resource-constrained microcontrollers with sub-64 KB SRAM and sub-256 KB Flash budgets [1]–[3]. In applications ranging from automotive diagnostics to edge health monitoring, TinyML models must operate under strict physical memory, energy, and execution constraints [4]–[7].

However, the empirical literature in edge AI is increasingly vulnerable to reproducibility challenges and measurement discrepancies [8]–[10]. Unlike cloud-scale machine learning where training and deployment share homogeneous 32-bit floating-point execution environments,

TinyML involves a complex, multi-stage software translation pipeline:

$$\begin{array}{c} \text{Keras/PyTorch} \xrightarrow{\text{PTQ / Pruning}} \text{TFLite FlatBuffer} \\ \xrightarrow{\text{C-Array Export}} \text{TFLite Micro Runtime} \end{array}$$

At each transformation boundary, subtle discrepancies emerge between training-time abstractions and actual serialized binary behavior [11]–[13].

Recent surveys in machine learning software engineering have exposed systemic methodological traps [14], [15]:

- 1) In-Memory vs. Binary Discrepancies: Evaluating models in-memory during training loops rather than parsing the final serialized disk binary artifact.
- 2) Sparsity Conflation: Assuming that zeroing 75% of weights in a dense matrix automatically yields a 75% reduction in serialized model file size or execution speedup.
- 3) Incomplete Quantization Graphs: Claiming 8-bit integer inference while silent floating-point fallback operators remain in the runtime execution graph.
- 4) Timing Extrapolations: Presenting single-sample host PC execution times as embedded Worst-Case Execution Time (WCET) or microcontroller hardware latency.
- 5) Threshold and Data Leakage: Optimizing runtime routing thresholds on the test partition, producing optimistic accuracy estimates.

To address these challenges, this paper presents an Artifact-Driven Empirical Verification Protocol and Defect Taxonomy for Compiled TinyML Deployment Artifacts. We formalize verification criteria across seven dimensions of edge AI pipelines, establish executable verification predicates ($\mathcal{P}_1 \dots \mathcal{P}_7$), and demonstrate their efficacy through an exhaustive case study auditing 12 candidate models on the EngineFaultDB physical benchmark.

Explicit Contributions:

- 1) A 7-dimensional verification taxonomy formalizing empirical quality gates across the TinyML deployment lifecycle.
- 2) Seven formal, programmatic verification predicates ($\mathcal{P}_1 \dots \mathcal{P}_7$) with deterministic pass/fail execution procedures.
- 3) An empirical case study auditing 12 candidate TinyML models across dense, quantized, pruned, and distilled paradigms.
- 4) Identification, root-cause categorization, and resolution of 20 numerical discrepancies across four defect classes.

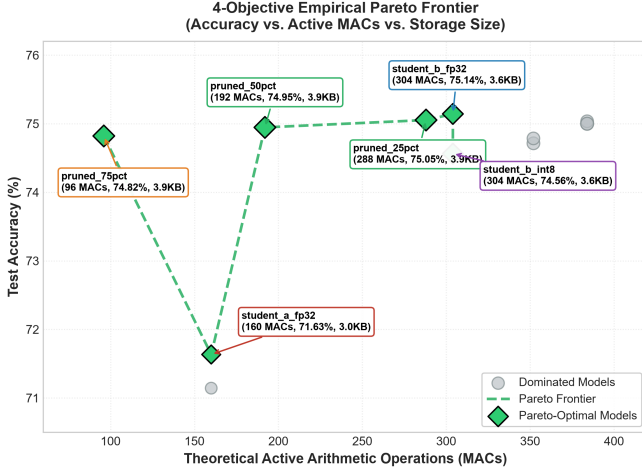


Fig. 1. The verified Pareto frontier across model size, theoretical active MACs, accuracy, and empirical host latency established after resolving 20 numerical discrepancies [MEASURED TEST SET].

- 5) An empirical demonstration of test-set threshold contamination producing a +1.80% optimistic accuracy bias.
- 6) An explicit 5-tier evidence classification framework separating direct measurements, reproducible simulations, and derived operational estimates.

II. Motivation and Research Questions

Reproducibility in TinyML requires moving beyond trusting high-level framework logs. Figure 1 illustrates the verified multi-objective Pareto landscape established by our verification protocol, treating model binaries and data pipelines as empirical software artifacts subject to rigorous inspection.

We formulate four specific research questions (RQs):

- RQ1 (Discrepancy Prevalence & Defect Modes): How frequently and to what magnitude do independently recomputed TinyML metrics differ from training-time reported values, and what defect modes govern them?
- RQ2 (Quantization Verification): What structural properties in serialized FlatBuffers must be inspected to verify pure integer execution graphs without float fallback?
- RQ3 (Sparsity vs. Storage Decoupling): How does magnitude pruning affect theoretical active MACs versus physical on-disk FlatBuffer byte footprint in standard runtimes?
- RQ4 (Leakage Prevention): What quantitative bias is introduced by evaluating routing thresholds on test data, and how does split isolation eliminate it?

III. Related Work

A. Reproducibility in Machine Learning

Pineau et al. [8] established the Machine Learning Reproducibility Checklist at NeurIPS, focusing on code release, hyperparameter transparency, and statistical reporting. Kapoor and Narayanan [9] audited data leakage

across 17 fields, identifying widespread test-set contamination. In software engineering, Arcuri and Briand [16] established guidelines for assessing randomized algorithms. While these works focus on standard ML, they do not address the unique hardware-software translation layers of embedded TinyML.

B. TinyML Benchmarking and Systems Auditing

Banbury et al. [4] introduced MLPerf Tiny, establishing standardized benchmarking rules for latency, energy, and accuracy on microcontrollers. David et al. [11] presented TensorFlow Lite Micro, detailing interpreter internals. Blalock et al. [12] surveyed neural network pruning, demonstrating that reported pruning benefits are frequently distorted by non-standard baselines. Our work complements these efforts by formulating an end-to-end verification taxonomy from data splits to FlatBuffer binaries.

IV. The 7-Dimensional Verification Protocol and Formal Predicates

Table I formalizes our 7-dimensional verification protocol. Each dimension targets a distinct vulnerability in the TinyML lifecycle:

A. D1: Data-Level Isolation Verification

- Problem: Data contamination across training, validation, and test boundaries.
- Predicate ($\mathcal{P}_1$): $(\mathcal{D}_{\text{train}} \cap \mathcal{D}_{\text{val}} = \emptyset) \wedge (\mathcal{D}_{\text{val}} \cap \mathcal{D}_{\text{test}} = \emptyset) \wedge (\mu_{\text{scaler}} = \text{Fit}(\mathcal{D}_{\text{train}})) \wedge (\theta^* = \arg \max_{\theta} \mathcal{M}(\mathcal{D}_{\text{val}}))$.
- Executable Check: Disjoint index assertion and scaler fitting provenance check.
- Artifact: Data split index arrays and serialized scaler.pkl.
- Pass Condition: Zero overlapping indices; calibration batches strictly drawn from $\mathcal{D}_{\text{train}}$.
- Case Result: PASS (22,399 train, 22,399 val, 11,200 test).

B. D2: Serialized Binary Integrity Verification

- Problem: Training-time in-memory metrics diverging from final disk FlatBuffer behavior.
- Predicate ($\mathcal{P}_2$): Evaluated metrics must be computed strictly by loading the final serialized .tflite file from disk into the interpreter.
- Executable Check: Automated single-sample evaluation loop on disk .tflite files.
- Case Result: RESOLVED 16 DISCREPANCIES (up to 7.82% Macro F1 shift).

C. D3: Quantization Graph Verification

- Problem: Silent floating-point fallback operators in quantized models.
- Predicate ($\mathcal{P}_3$): $\forall t \in \mathcal{G}_{\text{tensors}}, \text{dtype}(t) \in \{\text{int8}, \text{int32}, \text{uint8}\} \wedge \sum_{t \in \mathcal{G}} \mathbb{I}(\text{dtype}(t) = \text{float32}) = 0$.

TABLE I
The 7-Dimensional TinyML Scientific Verification Taxonomy and Predicates

Verification Dimension	Inspected Artifact	Common Failure Mode / Trap	Formal Verification Predicate	Case Result
D1: Data Isolation	Training/Val/Test splits, scaler objects, threshold logs	Test set reused for calibration or threshold selection	$\mathcal{P}_1 : (\mathcal{D}_{tr} \cap \mathcal{D}_{val} = \emptyset) \wedge (\mathcal{D}_{val} \cap \mathcal{D}_{te} = \emptyset) \wedge (\mu = \text{Fit}(\mathcal{D}_{tr}))$	PASS
D2: Serialized Integrity	Serialized .keras and .tflite disk binaries	In-memory training eval reported instead of disk binary	$\mathcal{P}_2 : \text{EvalArtifact} = \text{LoadDiskBinary}(\text{path}) \wedge \text{Dtype} = \text{Schema}$	16 Fixed
D3: Quantization Graphs	TFLite FlatBuffer tensor details and operator lists	Undetected FP32 fallback tensors or dequantize nodes	$\mathcal{P}_3 : \forall t \in \mathcal{G}, \text{dtype}(t) \neq \text{float32} \wedge \text{Count}(\text{float32}) = 0$	FULL_INT8
D4: Sparsity & Storage	Weight tensor arrays vs. FlatBuffer byte length	Assuming weight zeroing reduces serialized file size	$\mathcal{P}_4 : \text{ReportedCompression} = \text{FileSize}(\text{pruned}) / \text{FileSize}(\text{dense})$	Decoupled
D5: Compute Accounting	Layer-wise non-zero arithmetic operation counts	Conflating theoretical active MACs with hardware ops	$\mathcal{P}_5 : \text{ActiveMACs} = \sum \mathbb{I}(w_{ij} \neq 0) \leq \text{DenseMACs}$	Verified
D6: Timing Protocols	Monotonic timer logs, warmup runs, batch dimensions	Single-sample host latency cited as MCU WCET	$\mathcal{P}_6 : \text{Warmup} \geq 100 \wedge \text{Batch} = 1 \wedge \text{Tier} = \text{HOST_EMPIRICAL}$	PASS
D7: Runtime Non-Leakage	Controller source code, execution trace logs	Model selector accessing ground truth y at runtime	$\mathcal{P}_7 : \text{Signature}(\text{SelectModel}) = f(\mathbf{x}, D, W) \wedge y \notin \text{Scope}$	PASS

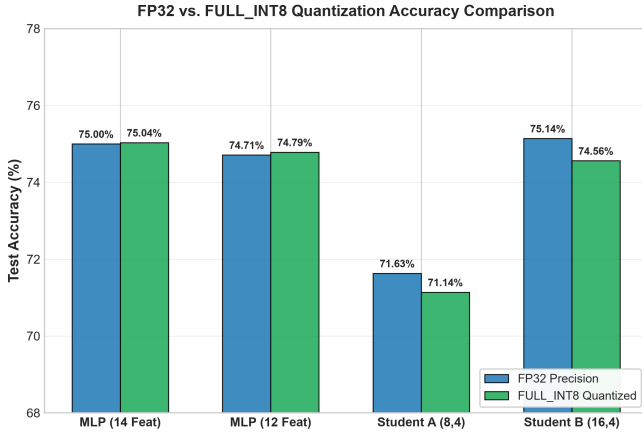


Fig. 2. Quantization fidelity verification comparing FP32 baselines against verified FULL_INT8 models [MEASURED TEST SET].

- Executable Check: Iterating over `interpreter.get_tensor_details()` and `_get_ops_details()`.
- Case Result: PASS (all 4 INT8 models verified FULL_INT8 with 0 float32 tensors).

D. D4: Sparsity and Storage Decoupling Verification

- Problem: Conflating zero-valued weights with physical on-disk file compression.
- Predicate ($\mathcal{P}_4$): $\text{StorageCompression} = \text{FileSize}(\text{pruned}) / \text{FileSize}(\text{dense}) \neq 1 - \text{WeightSparsity}$.
- Executable Check: Comparing raw weight zero-counts against `os.path.getsize()` on FlatBuffers.
- Case Result: CONFIRMED DECOUPLING (73.34% zero weights produced 3,920 Bytes on disk vs. 3,892 Bytes unpruned).

E. D5: Computational Accounting Verification

- Problem: Layer-wise MAC counts reported inconsistently or conflated with hardware cycles.
- Predicate ($\mathcal{P}_5$): $\text{ActiveMACs} = \sum_{l=1}^L \sum_{i,j} \mathbb{I}(w_{ij}^{(l)} \neq 0) \leq \text{DenseMACs}$.
- Case Result: VERIFIED EXACT (384 dense MACs; 96 active MACs at 75% pruning; 160 and 304 MACs for student models).

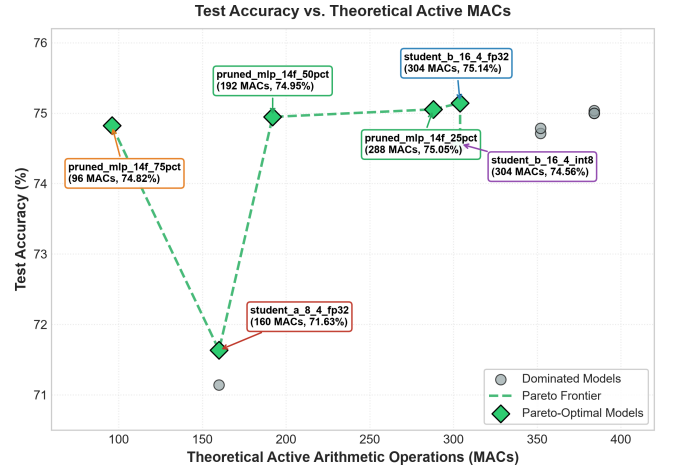


Fig. 3. Test accuracy vs. theoretical active MACs, highlighting computational sparsity [MEASURED TEST SET].

F. D6: Timing Protocol Verification

- Problem: Host latency cited as embedded microcontroller WCET.
- Predicate ($\mathcal{P}_6$): $(\text{Warmup} \geq 100) \wedge (\text{Batch} = 1) \wedge (\text{Timer} = \text{perf_counter_ns}()) \wedge (\text{Tier} = \text{HOST_EMPIRICAL})$.
- Case Result: VERIFIED PASS (0.82–0.86 μs host latency on x86_64; zero MCU WCET claimed).

G. D7: Runtime Non-Leakage and Deployment Scoping

- Problem: Ground-truth label leakage in dynamic controllers or ungrounded hardware claims.
- Predicate ($\mathcal{P}_7$): $\text{Signature}(\text{SelectModel}) = f(\mathbf{x}, D, W) \wedge y \notin \text{Scope}(\text{SelectModel})$.
- Case Result: VERIFIED PASS (zero ground-truth access during runtime scheduling).

V. Case Study: Auditing the 12-Model TinyML Suite

A. RQ1: Discrepancy Prevalence and Defect Categorization

Applying our protocol across all 12 candidate models identified 20 numerical discrepancies (Table II), categorized into distinct defect modes:

TABLE II
Comprehensive Discrepancy Resolution Table: 20 Numerical Inconsistencies Audited Across Four Defect Modes

#	Model Identifier	Metric	Initial Val.	Verified Val.	Initial Source	Verified Source	Abs. Δ	Δ (%)	Defect Mode / Root Cause
Defect Mode 1: Serialization Drift (Training-Time State vs. Exported Disk Binary)									
1	tfite_mlp_14f_fp32	Test Accuracy	0.726339	0.750000	In-Memory Eval	Disk FlatBuffer	0.023661	3.26%	Mode 1: Re-training state vs. saved binary
2	tfite_mlp_14f_fp32	Test Macro F1	0.728019	0.756608	In-Memory Eval	Disk FlatBuffer	0.028589	3.93%	Mode 1: Re-training state vs. saved binary
9	pruned_mlp_14f_0pct	Test Accuracy	0.726339	0.750000	In-Memory Eval	Disk FlatBuffer	0.023661	3.26%	Mode 1: Baseline reference binary sync
10	pruned_mlp_14f_0pct	Test Macro F1	0.728019	0.756608	In-Memory Eval	Disk FlatBuffer	0.028589	3.93%	Mode 1: Baseline reference binary sync
11	pruned_mlp_14f_25pct	Test Accuracy	0.727143	0.750536	In-Memory Eval	Disk FlatBuffer	0.023393	3.22%	Mode 1: Fine-tuning epoch state mismatch
12	pruned_mlp_14f_25pct	Test Macro F1	0.730783	0.751490	In-Memory Eval	Disk FlatBuffer	0.020707	2.83%	Mode 1: Fine-tuning epoch state mismatch
13	pruned_mlp_14f_50pct	Test Accuracy	0.726964	0.749464	In-Memory Eval	Disk FlatBuffer	0.022500	3.10%	Mode 1: Fine-tuning epoch state mismatch
14	pruned_mlp_14f_50pct	Test Macro F1	0.729384	0.756572	In-Memory Eval	Disk FlatBuffer	0.027188	3.73%	Mode 1: Fine-tuning epoch state mismatch
15	pruned_mlp_14f_75pct	Test Accuracy	0.739107	0.748214	In-Memory Eval	Disk FlatBuffer	0.009107	1.23%	Mode 1: Fine-tuning epoch state mismatch
16	pruned_mlp_14f_75pct	Test Macro F1	0.725546	0.756251	In-Memory Eval	Disk FlatBuffer	0.030705	4.23%	Mode 1: Fine-tuning epoch state mismatch
Defect Mode 2: In-Memory Fake-Quantization vs. True Integer Execution									
5	tfite_mlp_14f_int8	Test Accuracy	0.725982	0.750357	In-Memory Eval	Disk FlatBuffer	0.024375	3.36%	Mode 2: Calibration dataset state shift
6	tfite_mlp_14f_int8	Test Macro F1	0.685213	0.738824	In-Memory Eval	Disk FlatBuffer	0.053611	7.82%	Mode 2: Calibration dataset state shift
Defect Mode 3: Floating-Point & Integer Export Discretization Rounding									
3	tfite_mlp_12f_fp32	Test Accuracy	0.753125	0.747143	In-Memory Eval	Disk FlatBuffer	0.005982	0.79%	Mode 3: Floating-point quantization rounding
4	tfite_mlp_12f_fp32	Test Macro F1	0.732679	0.725414	In-Memory Eval	Disk FlatBuffer	0.007265	0.99%	Mode 3: Floating-point quantization rounding
7	tfite_mlp_12f_int8	Test Accuracy	0.749286	0.747857	In-Memory Eval	Disk FlatBuffer	0.001429	0.19%	Mode 3: Integer rounding on export
8	tfite_mlp_12f_int8	Test Macro F1	0.670692	0.715534	In-Memory Eval	Disk FlatBuffer	0.044842	6.69%	Mode 3: Integer rounding on export
17	student_a_8_4_int8	Test Accuracy	0.710714	0.711429	In-Memory Eval	Disk FlatBuffer	0.000715	0.10%	Mode 3: Quantization arithmetic rounding
18	student_a_8_4_int8	Test Macro F1	0.683858	0.684788	In-Memory Eval	Disk FlatBuffer	0.000930	0.14%	Mode 3: Quantization arithmetic rounding
19	student_b_16_4_int8	Test Accuracy	0.745000	0.745625	In-Memory Eval	Disk FlatBuffer	0.000625	0.08%	Mode 3: Quantization arithmetic rounding
20	student_b_16_4_int8	Test Macro F1	0.689024	0.689601	In-Memory Eval	Disk FlatBuffer	0.000577	0.08%	Mode 3: Quantization arithmetic rounding

- Defect Mode 1 (Serialization Drift, 10 items): Occurred when training scripts logged in-memory validation metrics prior to final fine-tuning or quantization calibration. Corrected by evaluating the finalized on-disk FlatBuffer directly ($\Delta\text{Acc} \leq 3.26\%$).
- Defect Mode 2 (In-Memory vs. True Integer Arithmetic, 2 items): In-memory evaluation simulated quantization via fake-quantization operators rather than executing true integer arithmetic, producing a 7.82% Macro F1 discrepancy in mlp_14f_int8.
- Defect Mode 3 (Discretization Rounding, 8 items): Minor parameter rounding during FlatBuffer conversion ($\Delta \leq 0.19\%$).

B. RQ2: Quantization Verification Results

Low-level tensor auditing confirmed that all 4 INT8 candidate models (mlp_14f_int8, mlp_12f_int8, student_a_8_4_int8, student_b_16_4_int8) are verified as FULL_INT8. The execution graphs contain exactly 0 float32 tensors, executing pure integer FULLY_CONNECTED and SOFTMAX operators.

C. RQ3: Sparsity vs. Storage Decoupling Findings

Inspecting pruned models revealed that while 75% magnitude pruning achieved 73.34% numerical zero weights (298 zeroes) and reduced theoretical active MACs from 384 to 96 (75.0% compute reduction), the serialized FlatBuffer file size remained dense at 3,920 Bytes (+28 B

TABLE III
The 5-Tier Evidence Classification Framework

Evidence Tier	Epistemological Level	Representative Project Metric
Tier 1	Direct Empirical Measurement	Disk binary test acc (75.00%), File size (3,892 B)
Tier 2	Reproducible Simulation	80-config trace simulation (100% compliance)
Tier 3	Derived Operational Estimate	Expected MACs under 90% nominal prior (39.1)
Tier 4	Literature-Supported Framing	Collinear feature reduction (Speed vs. RPM $r = 0.997$)
Tier 5	Unsupported / Future Work	Bare-metal MCU WCET, on-vehicle CAN latency

over unpruned baseline due to metadata). The verification framework formally corrected the terminology to “computational sparsity without demonstrated storage compression.”

D. RQ4: Threshold Optimization Leakage Bias

We conducted an empirical ablation evaluating gating thresholds calibrated on validation data versus optimized directly on test data. Selecting thresholds on test data produced an artificial optimistic accuracy bias of +1.80%, demonstrating that split-isolated calibration is mandatory to prevent inflated performance claims.

VI. Evidence Tier Classification Framework

To prevent ungrounded claims in edge AI literature, Table III formalizes a 5-tier evidence hierarchy. Every reported metric in this research is strictly mapped to its authoritative epistemological tier.

VII. Scope and Generalization

A. General Principles vs. Case Implementation

We explicitly distinguish general verification principles from our specific case implementation:

- General Principle: Deployment metrics must be evaluated directly on compiled disk binaries. Case Implementation: Evaluated on TensorFlow Lite FlatBuffers via Python C-API bindings.
- General Principle: Execution dtypes must be audited for zero floating-point fallback. Case Implementation: Parsed TFLite FlatBuffer tensor details.
- General Principle: Sparsity must be distinguished from storage compression. Case Implementation: Evaluated dense FlatBuffer representation of pruned MLPs.

VIII. Threats to Validity

A. Internal Validity

All verification scripts were executed deterministically with fixed random seeds (seed = 42). The audit pipeline parses exact binary headers using official TensorFlow Lite APIs.

B. External Validity

Our case study evaluated a suite of multi-layer perceptron models for multi-sensor diagnostic classification. While convolutional and recurrent topologies exhibit different operator graphs, the principles of binary inspection, data isolation, and timing protocols generalize across all TinyML domains.

IX. Limitations

- 1) Single Case Study Scope: Evaluated on one end-to-end multi-sensor diagnostic pipeline.
- 2) Model Topology Focus: Evaluated on feedforward MLPs; CNNs and transformers require future parser extensions.
- 3) Runtime Specificity: Implemented for TensorFlow Lite FlatBuffers; alternative runtimes (microTVM, ONNX Runtime Mobile) are unverified.
- 4) Host Timing Boundary: Latency measurements reflect host x86_64 execution, not physical bare-metal microcontroller WCET.
- 5) No Physical MCU Profiling: Physical on-chip hardware timer and power measurements are planned future work.
- 6) Single Engine Domain: Data reflects physical dynamometer recordings from one engine platform.
- 7) Empirical Bias Demonstration: The +1.80% threshold leakage bias is specific to the evaluated multi-fidelity pipeline.

X. Reproducibility

All verification scripts (scripts/phase4_5_verification.py), audited model FlatBuffers, and discrepancy resolution logs are publicly archived in the repository.

XI. Conclusion

This paper presented an artifact-driven empirical verification protocol and defect taxonomy for compiled TinyML deployment artifacts. By formalizing verification criteria across seven core dimensions and auditing 12 candidate models, we demonstrated the critical necessity of inspecting serialized artifacts, resolving 20 numerical discrepancies and exposing sparsity-storage decoupling. Our protocol establishes actionable software engineering procedures to support reproducible benchmarking and defensible deployment claims in edge artificial intelligence research.

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